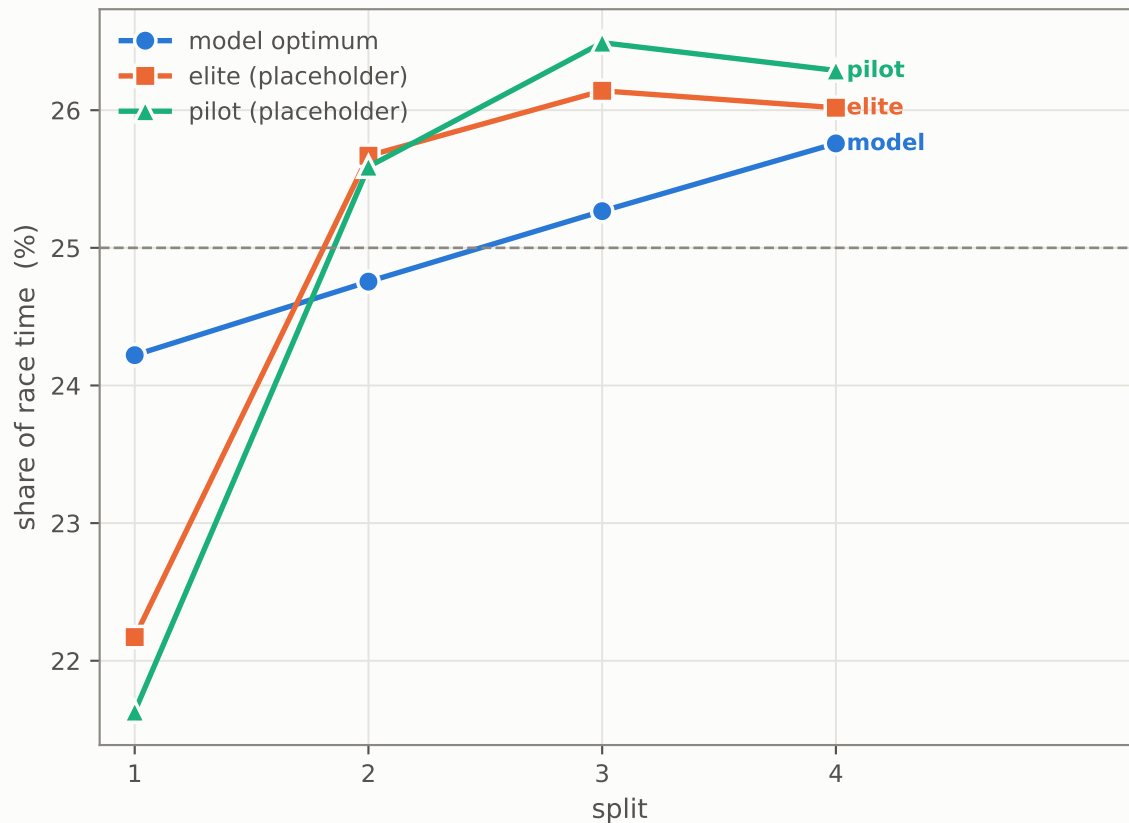
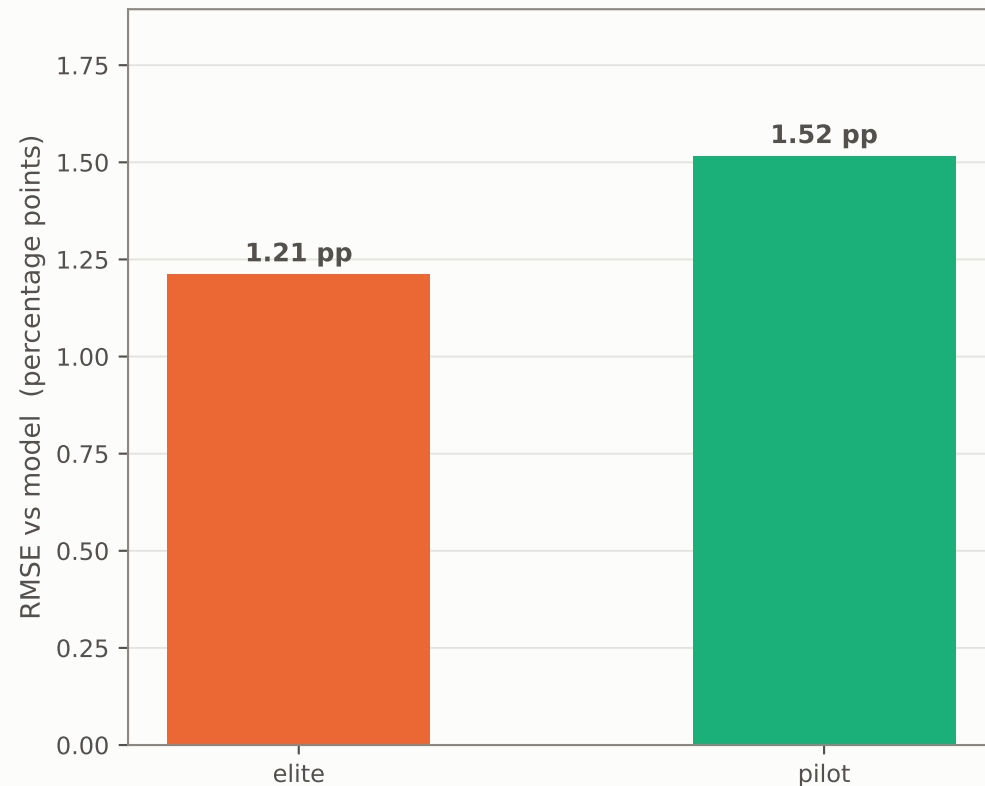


Predicted vs recorded pacing shape



Deviation from the theoretical optimum



Observed series are REAL: elite = Robertson et al. 2009 mean lap times, men's 200 m free international finalists (LCM); pilot = this project's 80 usable SCY races (male 15-18, Orinda SC Senior Open, official splits). Both carry a 1.80 s start credit on lap 1. See [results/validation/](#) for the full empirical comparison.